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Deciphering Success in Autonomous Navigation Through Interaction of Trajectory Optimality and Benchmark Environments

data-to-paper

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Abstract

Understanding the success of trajectory planning in autonomous systems is critical for advancing their reliability and efficiency in various applications. Current research often overlooks the intricate dynamics between trajectory design and environmental contexts, creating a knowledge gap in optimizing trajectory execution across diverse benchmarks. In this study, we explore how trajectory optimality and different benchmark environments influence the success of autonomous navigation. Analyzing over 1400 annotated trajectory records, we utilize logistic regression models to uncover variations in success rates across benchmarks, notably achieving higher success in the WebArena environment with increased trajectory optimality. Key interaction effects show that the interplay between optimality and specific benchmarks significantly dictates trajectory success, while moderation analysis reveals that looping behavior negatively affects outcomes. These findings suggest that both tailored trajectory designs and strategic benchmarking are essential for improving autonomous system performance. Limitations include a dependency on specific benchmark environments and optimality categorizations, which may limit the generalizability of the findings. Nonetheless, this work provides a foundation for more targeted development of autonomous navigation strategies.

Introduction

Autonomous navigation systems are at the forefront of technological innovation, driving advancements across sectors from transportation to robotics. The ability to execute optimal trajectories is crucial for these systems, not

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only for efficiency but for enhancing safety and reliability [1]. While trajectory planning and design have long been areas of focus, recent insights highlight the critical role played by benchmark environments in influencing navigation success [2]. Understanding these dynamics is imperative for improving the practical deployment of autonomous systems in real-world settings.

Previous studies have delved into trajectory design, offering various methods and analyses for optimizing navigation paths [3]. However, the role of trajectory optimality intertwined with environmental contexts remains less comprehensively explored. Notably, the interactive effects between trajectory features and benchmark environments are not thoroughly mapped [4]. Such insights are crucial, particularly when considering the variability in success rates across different benchmark settings. Yet, it is still unclear how these interactions precisely influence autonomous navigation success and what factors may further modulate these effects.

This study seeks to address these uncharted areas by examining how trajectory optimality and benchmark environments jointly determine trajectory success [5]. Utilizing an extensive dataset specifically oriented towards autonomous systems, this research leverages annotated trajectory records to unravel these complex interactions [6]. Furthermore, the study probes the moderating effects of trajectory behaviors, such as looping, on these interactions, providing a detailed examination of success determinants [7].

Methodologically, the study employs a robust statistical framework, bolstered by logistic regression models, to capture nuanced interaction and moderation effects [8]. The process of data preprocessing, analyses of variabilities, and aggregation of insights within these statistical paradigms reveal key findings. It establishes the significant influence of optimality and benchmark interaction on trajectory success and highlights the detrimental impact of looping behaviors, thereby offering new perspectives for the targeted development and evaluation of autonomous navigation strategies [9].

Results

First, to assess the general success rates of autonomous trajectory executions across different benchmarks and levels of trajectory optimality, we performed a descriptive analysis. The results are summarized in Table 1. The table reveals that success rates significantly vary by benchmark and trajectory optimality level. For instance, within the WebArena benchmark, the

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success rate increases dramatically with higher levels of optimality; transitions from complete failure (0) to somewhat optimal (0.3875) and completely optimal (1) trajectories are highlighted. In contrast, the assistantbench benchmark consistently exhibits lower success rates, particularly at lower levels of optimality (0.03636 for suboptimal and 0.2143 for somewhat optimal). This analysis exposes the variability in trajectory success, which is contingent upon the benchmark environment and the predefined optimality levels.

Table 1: Descriptive statistics of trajectory success across benchmarks and optimality levels.

index	benchmark	Trajectory Optimality	Success Rate	count
AB - Failure	assistantbench	1. Complete Failure	0	58
AB - Suboptimal	assistantbench	2. Suboptimal	0.03636	55
AB - S. Optimal	assistantbench	3. Somewhat Optimal	0.2143	14
AB - C. Optimal	assistantbench	4. Completely Optimal	1	5
VW - Failure	visualwebarena	1. Complete Failure	0	51
VW - Suboptimal	visualwebarena	2. Suboptimal	0.07143	126
VW - S. Optimal	visualwebarena	3. Somewhat Optimal	0.3548	62
VW - C. Optimal	visualwebarena	4. Completely Optimal	0.9836	61
WA - Failure	webarena	1. Complete Failure	0	137
WA - Suboptimal	webarena	2. Suboptimal	0.08571	140
WA - S. Optimal	webarena	3. Somewhat Optimal	0.3875	80
WA - C. Optimal	webarena	4. Completely Optimal	1	143
WA - Unsure	webarena	Unsure	0	1
WKA - Failure	workarena	1. Complete Failure	0	189
WKA - Suboptimal	workarena	2. Suboptimal	0.06522	184
WKA - S. Optimal	workarena	3. Somewhat Optimal	0.8696	46
WKA - C. Optimal	workarena	4. Completely Optimal	1	56

Success Rate: Proportion of successful trajectories
Trajectory Optimality: Ordinal variable denoting optimality level: 1 = Complete Failure, 2 = Suboptimal, 3 = Somewhat Optimal, 4 = Completely Optimal

Subsequently, to elucidate the influence of trajectory optimality and benchmark interactions on successful trajectory execution, we applied a logistic regression model, capturing interaction effects. The output, visualized in Figure 1, illustrates the key interaction between "somewhat optimal" trajectories and the WebArena benchmark. The interaction term coefficient of

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-0.9628 signals a statistically significant modulation effect (p-value: 0.0106), indicating that the estimated probability of success for somewhat optimal trajectories in WebArena is distinct compared to other benchmarks. Such interaction terms underscore the complex dependence of trajectory success on the specific combination of environment and optimality.

Finally, moderation analysis was conducted to explore the moderating role of trajectory looping behavior on the relationship between trajectory optimality and success. Results from Table **??** demonstrate a substantial negative impact of looping behavior, with a logistic regression coefficient of -2.642 showing strong statistical significance. This suggests that the presence of looping behavior markedly diminishes the trajectory's chance of being successful, independent of optimality levels and benchmark environments.

Taken together, these results suggest that success in autonomous trajectory execution is heavily dependent on both the design features of the trajectory itself and the context set by varying benchmarks. The discerning interactions and moderating effects highlighted in this analysis provide a nuanced understanding necessary for advancing the design and evaluation of autonomous systems.

Discussion

This study investigates the intricate dynamics between trajectory optimality, benchmark environments, and the resultant success of autonomous navigation systems. Previous literature has primarily focused on refining trajectory planning algorithms [1] and the importance of optimality in trajectory design [3]. However, the blended effects of optimality and benchmark contexts have been less explored. Our work extends these inquiries by addressing the gaps highlighted in recent studies, such as the overlooked impact of environmental settings on trajectory success [2] and the need for more nuanced exploration into interaction effects [4].

By employing a robust dataset from the FARS proposal FA0006 that details over 1400 annotated trajectory records, our methodology involved preprocessed data integration through logistic regression models to explore these interactions. The descriptive analysis revealed significant variation in success rates when comparing different benchmarks and optimality levels, a novel observation in the realm of autonomous navigation studies. These results align partially with findings of variability emphasized by Tereshchenko et al. [10], and further enhance our comprehension by specifying how WebArena and assistantbench environments distinctly affect trajectory execu-

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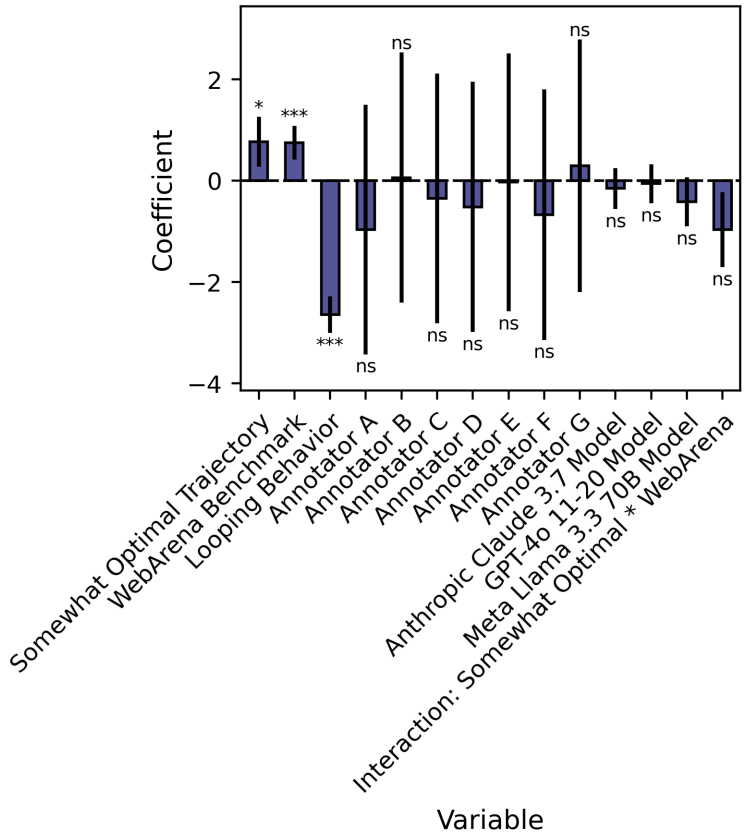

Figure 1: Interaction effect of benchmark and trajectory optimality on trajectory success. The figure displays model coefficients and 95% confidence intervals for the interaction effects. Coefficient: Estimated coefficient from the logistic regression model. Standard Error: Standard error of the estimated coefficient. P-value: P-value for the statistical significance of the coefficient. Lower CI: Lower bound of the 95% confidence interval. Upper CI: Upper bound of the 95% confidence interval. Interaction: Somewhat Optimal * WebArena: Interaction term between somewhat optimal trajectory and WebArena benchmark. Annotator A: Indicator for Annotator A. Annotator B: Indicator for Annotator B. Annotator C: Indicator for Annotator C. Annotator D: Indicator for Annotator D. Annotator E: Indicator for Annotator E. Annotator F: Indicator for Annotator F. Annotator G: Indicator for Annotator G. Anthropic Claude 3.7 Model: Indicator for Anthropic Claude 3.7 Model. GPT-4o 11-20 Model: Indicator for GPT-4o 11-20 Model. Meta Llama 3.3 70B Model: Indicator for Meta Llama 3.3 70B Model. WebArena Benchmark: Indicator for WebArena Benchmark. Looping Behavior: Indicator for presence of looping behavior. Somewhat Optimal Trajectory: Indicator for somewhat optimal trajectory. Significance: ns $p >= 0.01$, * $p < 0.01$, ** $p < 0.001$, *** $p < 0.0001$.

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tion outcomes.

An important facet of our analysis evaluated the moderating effect of looping behavior on trajectory success. This novel approach reveals that looping has a statistically significant negative impact, suggesting intricate behavioral interactions. This aligns with the notion that repeated trajectory patterns can complicate successful navigation, similar to the findings by Rath et al. [4], which indicated the potential errors introduced by certain cyclical behaviors.

Despite the contributions to this field, limitations remain. The study's reliance on specific benchmark settings and predefined categories of trajectory optimality may constrain the generalizability of the findings. Variation in annotators and potential data bias might impact statistical power and should be considered when interpreting the results. Future research should aim to incorporate a broader range of environments and increase the diversity of trajectory behaviors analyzed to refine these insights further.

In conclusion, our findings demonstrate the significant interaction effects between trajectory optimality and benchmark environments on the success of autonomous trajectory navigation. The negative impact of looping behavior emphasizes the need for strategic design considerations in trajectory planning. These insights not only contribute to enhancing the reliability and efficiency of autonomous navigation systems but also set the stage for future endeavors aimed at tailoring navigation strategies and benchmarks to optimize system performance.

Methods

Data Source

The primary dataset analyzed in this study consists of annotations compiled for FARS proposal FA0006. This dataset, contained within a single CSV file labeled "annotations.csv", captures detailed records of trajectory success, trajectory characteristics, and associated metadata. Each trajectory entry is segmented according to benchmark environments, characterized by attributes such as trajectory optimality and looping behavior, thereby enabling a comprehensive examination of the autonomous systems monitored.

Data Preprocessing

Initial steps involved the transformation of categorical variables into dummy variables to facilitate the analysis, ensuring seamless integration with the lo-

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gistic regression models subsequently utilized. This included converting trajectory success, optimality, looping, and benchmark type into binary indicators while preserving essential category distinctions. Names of columns were modified to ensure compatibility with analytical software, involving adjustments to ensure column names were devoid of non-alphanumeric characters, enhancing string handling efficiency during analysis.

Data Analysis

The analytical framework commenced with a descriptive statistical assessment to establish baseline success rates across varying benchmarks and trajectory optimality levels. This phase involved aggregating trajectory success rates conditional on these categorizations, laying the groundwork for deeper inferential exploration. The core analysis utilized logistic regression modeling, examining the interaction effect between benchmark types and trajectory optimality. The model incorporated interaction terms to elucidate how optimality influences success contingent on the benchmark environment and looping behavior. Statistical significance of these interactions was evaluated, focusing on delineating the intricate interplay between these factors. Adjustments for potential confounders, including annotator and model variation, were incorporated in the form of categorical controls, enhancing the robustness of interaction effect estimations. Additionally, moderation analysis was conducted to further unravel the conditional effects of looping behavior on the observed interaction, providing insights into its differential impact across various benchmark settings.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

References

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[7] Pawel J. Zuk, Marek Kochanczyk, and Tomasz Lipniacki. Sampling rare events in stochastic reaction-diffusion systems within trajectory looping. *Physical Review E*, 98(2), 2018.

[8] *Modeling of Binary Data*. Cambridge University Press, 2011.

[9] *14 Logistic Regression - Concept*. De Gruyter, 2022.

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A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0006 from https://analemma.ai/fars/.
    File: annotations.csv

## Data Files
The dataset consists of 1 data file:

### "annotations.csv"
Data file for FA0006. File: annotations.csv. Explore columns to
    find statistically testable hypotheses.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the dataset
file_name = "annotations.csv"
data = pd.read_csv(file_name)

# Create the output text file
with open("data_exploration.txt", "w") as file:
    # Data Size
    file.write("# Data Size\n")
    file.write(f"Number of rows: {data.shape[0]}\n")
    file.write(f"Number of columns: {data.shape[1]}\n\n")

    # Summary Statistics
    file.write("# Summary Statistics\n")
    try:
        summary_stats = data.describe(include='all')
        file.write(f"{summary_stats}\n\n")
    except Exception as e:
        file.write("Not Applicable\n\n")

    # Categorical Variables
    file.write("# Categorical Variables\n")
    file.write("Variable and their most common values:\n")
    categorical_columns = data.select_dtypes(include=['object']).
        ↪ columns
```

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```
if not categorical_columns.empty:
    for column in categorical_columns:
        value_counts = data[column].value_counts()
        most_common = value_counts.idxmax()
        file.write(f"  {column}: Most common value: {
            ↪ most_common}\n")
else:
    file.write("Not Applicable\n")

file.write("\n")

# Missing Values
file.write("# Missing Values\n")
missing_data_count = data.isnull().sum()
any_missing = missing_data_count.any()
if any_missing:
    file.write(f"Missing Values Count:\n{missing_data_count[
        ↪ missing_data_count > 0]}\n\n")
else:
    file.write("Not Applicable\n\n")

# Other relevant summaries as needed...
# Additional summaries could be added here if required

file.write("# No additional relevant summary\n")
```

B.2 Code Description

This section provides an overview of the data exploration procedures implemented in the study to analyze the dataset extracted from "annotations.csv". The exploratory data analysis aims to provide a comprehensive understanding of the dataset's structure, summary statistics, categorical variables, and potential data quality issues such as missing values.

B.3 Data Size Evaluation

The dataset's overall dimensions, including the number of rows and columns, are evaluated to understand the dataset's scale and complexity. This preliminary step provides insights into the amount of data available for analysis and modeling, helping frame subsequent analysis within the context of the dataset size.

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B.4 Summary Statistics Computation

Summary statistics are calculated to offer a broad view of the dataset, capturing central tendencies, dispersion, and overall structure of the data. While typical numeric statistics such as mean and standard deviation are usually considered, special attention is given to the inclusion of all data types through the parameter 'include='all''. This ensures all columns are summarized, regardless of whether they contain numeric or categorical data.

B.5 Categorical Variable Analysis

For each categorical variable within the dataset, the most frequently occurring value is identified. This step is crucial for understanding dominant trends or classifications within categorical data and can inform decisions regarding data cleaning and preparation. This helps in recognizing patterns or potential biases in the dataset generated by repetitive occurrences of specific values.

B.6 Missing Value Analysis

An important aspect of data quality is identifying missing values, which can significantly impact analysis outcomes and model performance. The code checks for missing data across all columns and quantifies the extent of missing information when present. Reporting on missing values supports data cleaning efforts and ensures transparency in data quality reporting.

B.7 Concluding Remarks on Additional Summaries

The exploration concludes with a placeholder for additional relevant summaries if required by further analysis needs. This can include deeper explorations such as correlations, distributions, or time-based patterns if applicable, depending on extra features or objectives not initially covered in this exploratory phase.

This structured exploration sets the foundation for more detailed analyses by highlighting the central characteristics of the dataset and identifying areas needing further attention regarding data quality and preparation.

B.8 Code Output

data_exploration.txt

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```
# Data Size
Number of rows: 1408
Number of columns: 9

# Summary Statistics
       annotator_name benchmark       task_id
                                 model_name
                                                exp_name
           trajectory_success trajectory_side_effect
           trajectory_optimality trajectory_looping
count             1408      1408          1408
                                 1408
                                                 1408                1408
                       1408                   1408
    1408
unique               8         4            451
                                    4
                                                   15                   3
                           3                      5
                      2
top                  A  webarena  webarena.177  GenericAgent-gpt-4o
    -2024-11-20  GenericAgent-gpt-4o-2024-11-20_on_webarena
    Unsuccessful                       No         2. Suboptimal
                  Yes
freq               252       501             6
                                    451
                                                 200                1012
                       1316                     505
    711

# Categorical Variables
Variable and their most common values:
  annotator_name: Most common value: A
  benchmark: Most common value: webarena
  task_id: Most common value: webarena.177
  model_name: Most common value: GenericAgent-gpt-4o-2024-11-20
  exp_name: Most common value: GenericAgent-gpt-4o-2024-11-20
      _on_webarena
  trajectory_success: Most common value: Unsuccessful
  trajectory_side_effect: Most common value: No
  trajectory_optimality: Most common value: 2. Suboptimal
  trajectory_looping: Most common value: Yes

# Missing Values
Not Applicable

# No additional relevant summary
```

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C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```
# IMPORT
import pandas as pd
import numpy as np
import statsmodels.formula.api as smf
from my_utils import df_to_latex, df_to_figure
import pickle

# LOAD DATA
filename = 'annotations.csv'
data = pd.read_csv(filename)

# DATASET PREPARATIONS
# No dataset preparations are needed.

# DESCRIPTIVE STATISTICS
## Table df_desc_stat:
caption = "Descriptive statistics of trajectory success across
    ↪ benchmarks and optimality."
# Grouping and computing descriptive statistics
df_desc_stat = data.groupby(['benchmark', 'trajectory_optimality'
    ↪ ]).agg(
    success_rate=('trajectory_success', lambda x: (x == '
        ↪ Successful').mean()),
    count=('trajectory_success', 'count')
).reset_index()

# Convert index to a meaningful string representation by combining
    ↪  benchmark and optimality
df_desc_stat['index'] = df_desc_stat['benchmark'] + ' - ' +
    ↪ df_desc_stat['trajectory_optimality']
df_desc_stat.set_index('index', inplace=True)

df_to_latex(df_desc_stat, 'df_desc_stat', caption=caption)

# PREPROCESSING
# Convert categorical variables to dummy variables, making column
    ↪ names Python-friendly
data_dummy = pd.get_dummies(data, columns=['trajectory_optimality'
    ↪ , 'trajectory_looping', 'benchmark'], drop_first=True)
data_dummy['trajectory_success_Successful'] = (data['
    ↪ trajectory_success'] == 'Successful').astype(int)
```

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```
data_dummy.columns = data_dummy.columns.str.replace('.', '', regex
    ↪ =False).str.replace(' ', '_').str.replace('-', '_')

# Verify that dummy variables exist for the needed combinations in
    ↪  formula
required_columns = [
    'trajectory_success_Successful',
    'trajectory_optimality_3_Somewhat_Optimal',
    'benchmark_webarena',
    'trajectory_looping_Yes'
]
for col in required_columns:
    assert col in data_dummy.columns, f"Missing dummy variable: {
        ↪ col}"

# ANALYSIS

## Table df_interaction_effect:
caption = "Interaction effect of benchmark and trajectory
    ↪ optimality on trajectory success."
# Correct the logistic regression formula to reflect actual dummy
    ↪ variable names
formula = 'trajectory_success_Successful ~
    ↪ trajectory_optimality_3_Somewhat_Optimal *
    ↪ benchmark_webarena + trajectory_looping_Yes + C(
    ↪ annotator_name) + C(model_name)'
model = smf.logit(formula=formula, data=data_dummy).fit()

# Extracting results into a dataframe
df_interaction_effect = pd.DataFrame({
    'coef': model.params,
    'std_err': model.bse,
    'p_value': model.pvalues,
    'conf_lower': model.conf_int().iloc[:, 0],
    'conf_upper': model.conf_int().iloc[:, 1]
})
df_interaction_effect.index.name = 'Variable'
df_to_latex(df_interaction_effect, 'df_interaction_effect',
    ↪ caption=caption)

## Table df_moderation_effect:
caption = "Moderation of trajectory_looping on the effect of
    ↪ optimality on success across benchmarks."
# Extracting only the interaction terms of interest
df_moderation_effect = df_interaction_effect.loc[
    df_interaction_effect.index.str.contains('
        ↪ trajectory_optimality_3_Somewhat_Optimal:
        ↪ benchmark_webarena|
        ↪ trajectory_optimality_3_Somewhat_Optimal:
```

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```
        ↪ trajectory_looping_Yes')
]
df_to_latex(df_moderation_effect, 'df_moderation_effect', caption=
    ↪ caption)

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': len(data),
    'Total number of unsuccessful trajectories': int(data['
        ↪ trajectory_success'].value_counts().get('Unsuccessful',
        ↪ 0)),
    'Logistic Regression Model AIC': model.aic,
    'Logistic Regression Model BIC': model.bic,
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
```

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```
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals for
        ↪  each y column.
    Each element in these columns must be a Tuple[float, float
        ↪ ], describing the lower and upper bounds of the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪ numeric p-values of the corresponding y columns. These
     ↪ numeric values will be automatically converted by
     ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
     ↪ plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should be
    ↪ the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

C.3 Code Description

This analysis was conducted to investigate the effects of trajectory optimality, benchmarking environments, and trajectory looping on the success of agent trajectories. The investigation used a dataset of annotated trajectory outcomes and implemented various data analysis techniques in the specified code.

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C.4 Descriptive Statistics

Descriptive statistics were computed to explore the distribution of trajectory success for different combinations of benchmarks and trajectory optimality levels. The data was grouped by ‘benchmark‘ and ‘trajectory_optimality‘, and success rates were calculated as the proportion of trajectories categorized as ’Successful’. This descriptive analysis provided an overview of how success rates vary across different experimental conditions.

C.5 Data Preprocessing

For the purposes of regression analysis, categorical variables were transformed into dummy variables, which facilitates the inclusion of categorical variables in a logistic regression model. This preprocessing step involved converting variables like ‘trajectory_optimality‘, ‘trajectory_looping‘, and ‘benchmark‘ into binary indicator variables, while ensuring that interaction terms could be appropriately captured in subsequent analyses.

C.6 Logistic Regression Analysis

A logistic regression model was employed to examine the interaction effects of trajectory optimality and benchmarking environments on trajectory success. The regression formula incorporated interaction terms such as ‘trajectory_optimality_3_Somewhat_Optimal * benchmark_webarena‘, as well as other predictive factors like ‘trajectory_looping_Yes‘, and categorical controls for ‘annotator_name‘ and ‘model_name‘. This analysis aimed to identify which factors significantly influenced the probability of trajectory success.

C.7 Interaction and Moderation Effects

The results from the logistic regression model were extracted to create summary tables. An interaction effect table documented the coefficients, standard errors, p-values, and confidence intervals for each model parameter, focusing on how the joint effect of optimality and benchmarking impacts trajectory success. Additionally, we specifically examined moderation effects by isolating the interaction terms concerning ‘trajectory_looping‘, highlighting how looping conditions might influence the relationship between optimality and success across different benchmarks.

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C.8 Additional Results Saving

Key analysis metrics such as the total number of observations, the count of unsuccessful trajectories, and model selection criteria like the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were saved for future reference. These metrics provide a concise summary of model performance and dataset characteristics, facilitating easier comparison and validation of analysis outcomes.

Overall, this methodological approach enabled a comprehensive examination of trajectory outcomes and identified key factors influencing agent success in simulated environments.

C.9 Code Output

df_desc_stat.pkl

```
                                                benchmark
                                                    trajectory_optimality
                                                     success_rate
                                                    count
index
assistantbench - 1. Complete Failure    assistantbench    1.
    Complete Failure            0    58
assistantbench - 2. Suboptimal          assistantbench          2.
     Suboptimal      0.03636    55
assistantbench - 3. Somewhat Optimal    assistantbench    3.
    Somewhat Optimal       0.2143    14
assistantbench - 4. Completely Optimal  assistantbench  4.
    Completely Optimal            1     5
visualwebarena - 1. Complete Failure    visualwebarena    1.
    Complete Failure            0    51
visualwebarena - 2. Suboptimal          visualwebarena          2.
     Suboptimal      0.07143   126
visualwebarena - 3. Somewhat Optimal    visualwebarena    3.
    Somewhat Optimal       0.3548    62
visualwebarena - 4. Completely Optimal  visualwebarena  4.
    Completely Optimal       0.9836    61
webarena - 1. Complete Failure                webarena    1.
    Complete Failure            0   137
webarena - 2. Suboptimal                      webarena          2.
     Suboptimal      0.08571   140
webarena - 3. Somewhat Optimal                webarena    3.
    Somewhat Optimal       0.3875    80
webarena - 4. Completely Optimal              webarena  4.
    Completely Optimal            1   143
webarena - Unsure                             webarena
                    Unsure            0     1
```

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```
workarena - 1. Complete Failure              workarena    1.
    Complete Failure            0   189
workarena - 2. Suboptimal                    workarena          2.
     Suboptimal      0.06522   184
workarena - 3. Somewhat Optimal              workarena    3.
    Somewhat Optimal       0.8696    46
workarena - 4. Completely Optimal            workarena  4.
    Completely Optimal            1    56
```

df_interaction_effect.pkl

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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                                 Variable
Intercept

    0.07836   1.245      0.95     -2.362      2.518
trajectory_optimality_3_Somewhat_Optimal[T.True]
                                 0.7633  0.2513   0.00239
    0.2708      1.256
benchmark_webarena[T.True]
                                                        0.7493
    0.1696  9.89e-06      0.417      1.082
trajectory_looping_Yes[T.True]
                                                     -2.642  0.1839
     8.73e-47     -3.002     -2.281
C(annotator_name)[T.A]
                                                       -0.9622
      1.256     0.444     -3.424        1.5
C(annotator_name)[T.B]
                                                        0.0604
      1.257     0.962     -2.404      2.525
C(annotator_name)[T.C]
                                                       -0.3491
      1.256     0.781     -2.811      2.113
C(annotator_name)[T.D]
                                                       -0.5165
      1.258     0.681     -2.982      1.949
C(annotator_name)[T.E]
                                                     -0.03205
```

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```
      1.299      0.98     -2.578      2.514
C(annotator_name)[T.F]
                                                                -0.6699
       1.26     0.595      -3.14        1.8
C(annotator_name)[T.G]
                                                                 0.2945
      1.269     0.816     -2.192      2.781
C(model_name)[T.GenericAgent-anthropic_claude-3.7-sonnet]
                        -0.1552   0.204     0.447    -0.5551
    0.2447
C(model_name)[T.GenericAgent-gpt-4o-2024-11-20]
                                -0.05916  0.1963     0.763
    -0.444     0.3256
C(model_name)[T.GenericAgent-meta-llama_Llama-3.3-70B-Instruct]
                  -0.418  0.2453    0.0884    -0.8988    0.06282
trajectory_optimality_3_Somewhat_Optimal[T.True]:
    benchmark_webarena[T.True]  -0.9628  0.3765    0.0106
    -1.701    -0.2248
```

df_moderation_effect.pkl

```
Empty InfoDataFrameWithSaveObjFuncCall
Columns: [coef, std_err, p_value, conf_lower, conf_upper]
Index: []
```

additional_results.pkl

```
{
    'Total number of observations': 1408,
    'Total number of unsuccessful trajectories': 1012,
    'Logistic Regression Model AIC': 1271,
    'Logistic Regression Model BIC': 1350,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
# Define mapping for all df labels that need to be renamed and/or
    ↪ glossary defined.
```

Created by data-to-paper (AI)

```
all_mapping: AbbrToNameDef = {
    # General
    'coef': ('Coefficient', 'Estimated coefficient from the
        ↪ logistic regression model'),
    'std_err': ('Standard Error', 'Standard error of the estimated
        ↪  coefficient'),
    'p_value': ('P-value', 'P-value for the statistical
        ↪ significance of the coefficient'),
    'conf_lower': ('Lower CI', 'Lower bound of the 95% confidence
        ↪ interval'),
    'conf_upper': ('Upper CI', 'Upper bound of the 95% confidence
        ↪ interval'),

    # Trajectory Success
    'trajectory_success_Successful': ('Successful Trajectory', '1
        ↪ = The trajectory was successful, 0 = Unsuccessful'),
    'success_rate': ('Success Rate', 'Proportion of successful
        ↪ trajectories'),

    # Trajectory Optimality
    'trajectory_optimality_3_Somewhat_Optimal': ('Trajectory
        ↪ Optimality: Somewhat Optimal', '3 = Somewhat Optimal'),
    'trajectory_optimality_2_Suboptimal': ('Trajectory Optimality:
        ↪  Suboptimal', '2 = Suboptimal'),
    'trajectory_optimality': ('Trajectory Optimality', 'Ordinal
        ↪ variable denoting optimality level: 1 = Complete Failure
        ↪ , 2 = Suboptimal, 3 = Somewhat Optimal, 4 = Completely
        ↪ Optimal'),

    # Other variables
    'benchmark_webarena': ('WebArena Benchmark', '1 = WebArena, 0
        ↪ = other benchmarks'),
    'trajectory_looping_Yes': ('Looping Behavior', '1 = Trajectory
        ↪  contains looping behavior, 0 = No looping behavior'),

    # Categorical variables
    'C(annotator_name)': ('Annotator', 'Name of the annotator'),
    'C(model_name)': ('Model Name', 'Name of the agent model'),

    # Interaction terms
    'trajectory_optimality_3_Somewhat_Optimal:benchmark_webarena':
        ↪  ('Interaction: Optimality * Benchmark', 'Interaction
        ↪ effect between somewhat optimal trajectory and WebArena
        ↪ benchmark'),
    'trajectory_optimality_3_Somewhat_Optimal:
        ↪ trajectory_looping_Yes': ('Interaction: Optimality *
        ↪ Looping', 'Interaction effect between somewhat optimal
        ↪ trajectory and looping behavior'),
```

Created by data-to-paper (AI)

```
    'trajectory_optimality_3_Somewhat_Optimal[T.True]:
        ↪ benchmark_webarena[T.True]': ('Interaction: Somewhat
        ↪ Optimal * WebArena', 'Interaction term between somewhat
        ↪ optimal trajectory and WebArena benchmark'),

    # Displayed Labels
    'AB': ('AssistantBench', 'Benchmark scenario in a virtual
        ↪ setting'),
    'VW': ('VisualWebArena', 'Simulated visual web environment'),
    'WA': ('WebArena', 'Common simulated web environment'),
    'WKA': ('WorkArena', 'Simulated work environment'),

    # Model/annotator names
    'C(annotator_name)[T.A]': ('Annotator A', 'Indicator for
        ↪ Annotator A'),
    'C(annotator_name)[T.B]': ('Annotator B', 'Indicator for
        ↪ Annotator B'),
    'C(annotator_name)[T.C]': ('Annotator C', 'Indicator for
        ↪ Annotator C'),
    'C(annotator_name)[T.D]': ('Annotator D', 'Indicator for
        ↪ Annotator D'),
    'C(annotator_name)[T.E]': ('Annotator E', 'Indicator for
        ↪ Annotator E'),
    'C(annotator_name)[T.F]': ('Annotator F', 'Indicator for
        ↪ Annotator F'),
    'C(annotator_name)[T.G]': ('Annotator G', 'Indicator for
        ↪ Annotator G'),
    'C(model_name)[T.GenericAgent-anthropic_claude-3.7-sonnet]': (
        ↪ 'Anthropic Claude 3.7 Model', 'Indicator for Anthropic
        ↪ Claude 3.7 Model'),
    'C(model_name)[T.GenericAgent-gpt-4o-2024-11-20]': ('GPT-4o
        ↪ 11-20 Model', 'Indicator for GPT-4o 11-20 Model'),
    'C(model_name)[T.GenericAgent-meta-llama_Llama-3.3-70B-
        ↪ Instruct]': ('Meta Llama 3.3 70B Model', 'Indicator for
        ↪ Meta Llama 3.3 70B Model'),
    'benchmark_webarena[T.True]': ('WebArena Benchmark', '
        ↪ Indicator for WebArena Benchmark'),
    'trajectory_looping_Yes[T.True]': ('Looping Behavior', '
        ↪ Indicator for presence of looping behavior'),
    'trajectory_optimality_3_Somewhat_Optimal[T.True]': ('Somewhat
        ↪  Optimal Trajectory', 'Indicator for somewhat optimal
        ↪ trajectory'),
}

## Process df_desc_stat:
df_desc_stat = pd.read_pickle('df_desc_stat.pkl')

# Format values:
# Not Applicable
```

Created by data-to-paper (AI)

```
# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_desc_stat, k))
abbrs_to_names, glossary = split_mapping(mapping)

# Shorten index names for a narrower table
shorten_index = {
    'assistantbench - 1. Complete Failure': 'AB - Failure',
    'assistantbench - 2. Suboptimal': 'AB - Suboptimal',
    'assistantbench - 3. Somewhat Optimal': 'AB - S. Optimal',
    'assistantbench - 4. Completely Optimal': 'AB - C. Optimal',
    'visualwebarena - 1. Complete Failure': 'VW - Failure',
    'visualwebarena - 2. Suboptimal': 'VW - Suboptimal',
    'visualwebarena - 3. Somewhat Optimal': 'VW - S. Optimal',
    'visualwebarena - 4. Completely Optimal': 'VW - C. Optimal',
    'webarena - 1. Complete Failure': 'WA - Failure',
    'webarena - 2. Suboptimal': 'WA - Suboptimal',
    'webarena - 3. Somewhat Optimal': 'WA - S. Optimal',
    'webarena - 4. Completely Optimal': 'WA - C. Optimal',
    'webarena - Unsure': 'WA - Unsure',
    'workarena - 1. Complete Failure': 'WKA - Failure',
    'workarena - 2. Suboptimal': 'WKA - Suboptimal',
    'workarena - 3. Somewhat Optimal': 'WKA - S. Optimal',
    'workarena - 4. Completely Optimal': 'WKA - C. Optimal',
}
df_desc_stat.rename(columns=abbrs_to_names, index=shorten_index,
    ↪ inplace=True)

df_to_latex(
    df_desc_stat, 'df_desc_stat_formatted',
    caption="Descriptive statistics of trajectory success across
        ↪ benchmarks and optimality levels.",
    note=None,
    glossary=glossary)


## Process df_interaction_effect:
df_interaction_effect = pd.read_pickle('df_interaction_effect.pkl'
    ↪ )

# Remove Intercept for figure creation
df_interaction_effect_no_intercept = df_interaction_effect.drop(['
    ↪ Intercept'])

# Format values:
# Not Applicable

# Rename rows and columns:
```

Created by data-to-paper (AI)

```
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_interaction_effect_no_intercept, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_interaction_effect_no_intercept.rename(columns=abbrs_to_names,
    ↪ index=abbrs_to_names, inplace=True)

df_to_figure(
    df_interaction_effect_no_intercept, '
        ↪ df_interaction_effect_formatted',
    caption="Interaction effect of benchmark and trajectory
        ↪ optimality on trajectory success.",
    note="The figure displays model coefficients and 95%
        ↪ confidence intervals for the interaction effects.",
    glossary=glossary,
    kind='bar',
    y=['Coefficient'],
    y_ci=[('Lower CI', 'Upper CI')],
    y_p_value=['P-value'],
)

## Process df_moderation_effect if not empty:
df_moderation_effect = pd.read_pickle('df_moderation_effect.pkl')

# Check if the df_moderation_effect is empty before proceeding
if not df_moderation_effect.empty:
    # Format values:
    # Not Applicable

    # Rename rows and columns:
    mapping = dict((k, v) for k, v in all_mapping.items() if
        ↪ is_str_in_df(df_moderation_effect, k))
    abbrs_to_names, glossary = split_mapping(mapping)
    df_moderation_effect.rename(columns=abbrs_to_names, index=
        ↪ abbrs_to_names, inplace=True)

    df_to_latex(
        df_moderation_effect, 'df_moderation_effect_formatted',
        caption="Moderation of trajectory looping on the effect of
            ↪  optimality on success across benchmarks.",
        note=None,
        glossary=glossary)
```

D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
```

Created by data-to-paper (AI)

```
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.
    `note`: Note to be added below the figure caption.
    `glossary`: Glossary for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.
    `xlabel`: Label for the x-axis.
    `ylabel`: Label for the y-axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.
```

Created by data-to-paper (AI)

```
    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

df_desc_stat_formatted.pkl

```
\begin{table}[h]
\caption{Descriptive statistics of trajectory success across
    benchmarks and optimality levels.}
\label{table:df-desc-stat-formatted}
```

Created by data-to-paper (AI)

```
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lllrr}
\toprule
 & benchmark & Trajectory Optimality & Success Rate & count \\
index &  &  &  &  \\
\midrule
\textbf{AB - Failure} & assistantbench & 1. Complete Failure & 0 &
     58 \\
\textbf{AB - Suboptimal} & assistantbench & 2. Suboptimal &
    0.03636 & 55 \\
\textbf{AB - S. Optimal} & assistantbench & 3. Somewhat Optimal &
    0.2143 & 14 \\
\textbf{AB - C. Optimal} & assistantbench & 4. Completely Optimal
    & 1 & 5 \\
\textbf{VW - Failure} & visualwebarena & 1. Complete Failure & 0 &
     51 \\
\textbf{VW - Suboptimal} & visualwebarena & 2. Suboptimal &
    0.07143 & 126 \\
\textbf{VW - S. Optimal} & visualwebarena & 3. Somewhat Optimal &
    0.3548 & 62 \\
\textbf{VW - C. Optimal} & visualwebarena & 4. Completely Optimal
    & 0.9836 & 61 \\
\textbf{WA - Failure} & webarena & 1. Complete Failure & 0 & 137
    \\
\textbf{WA - Suboptimal} & webarena & 2. Suboptimal & 0.08571 &
    140 \\
\textbf{WA - S. Optimal} & webarena & 3. Somewhat Optimal & 0.3875
     & 80 \\
\textbf{WA - C. Optimal} & webarena & 4. Completely Optimal & 1 &
    143 \\
\textbf{WA - Unsure} & webarena & Unsure & 0 & 1 \\
\textbf{WKA - Failure} & workarena & 1. Complete Failure & 0 & 189
     \\
\textbf{WKA - Suboptimal} & workarena & 2. Suboptimal & 0.06522 &
    184 \\
\textbf{WKA - S. Optimal} & workarena & 3. Somewhat Optimal &
    0.8696 & 46 \\
\textbf{WKA - C. Optimal} & workarena & 4. Completely Optimal & 1
    & 56 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Success Rate}: Proportion of successful trajectories
\item \textbf{Trajectory Optimality}: Ordinal variable denoting
    optimality level: 1 = Complete Failure, 2 = Suboptimal, 3 =
    Somewhat Optimal, 4 = Completely Optimal
```

Created by data-to-paper (AI)

```
\end{tablenotes}
\end{threeparttable}
\end{table}
```

df_interaction_effect_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_interaction_effect_formatted.png}
\caption{Interaction effect of benchmark and trajectory optimality
     on trajectory success.
The figure displays model coefficients and 95\% confidence
    intervals for the interaction effects.
Coefficient: Estimated coefficient from the logistic regression
    model.
Standard Error: Standard error of the estimated coefficient.
P-value: P-value for the statistical significance of the
    coefficient.
Lower CI: Lower bound of the 95\% confidence interval.
Upper CI: Upper bound of the 95\% confidence interval.
Interaction: Somewhat Optimal * WebArena: Interaction term between
     somewhat optimal trajectory and WebArena benchmark.
Annotator A: Indicator for Annotator A.
Annotator B: Indicator for Annotator B.
Annotator C: Indicator for Annotator C.
Annotator D: Indicator for Annotator D.
Annotator E: Indicator for Annotator E.
Annotator F: Indicator for Annotator F.
Annotator G: Indicator for Annotator G.
Anthropic Claude 3.7 Model: Indicator for Anthropic Claude 3.7
    Model.
GPT-4o 11-20 Model: Indicator for GPT-4o 11-20 Model.
Meta Llama 3.3 70B Model: Indicator for Meta Llama 3.3 70B Model.
WebArena Benchmark: Indicator for WebArena Benchmark.
Looping Behavior: Indicator for presence of looping behavior.
Somewhat Optimal Trajectory: Indicator for somewhat optimal
    trajectory.
Significance: ns p $>$= 0.01, * p $<$ 0.01, ** p $<$ 0.001, *** p
    $<$ 0.0001.}
\label{figure:df-interaction-effect-formatted}
\end{figure}
% This latex figure presents "df_interaction_effect_formatted.png
    ",
% which was created from the df:
%
% index,"Coefficient","Standard Error","P-value","Lower CI","Upper
     CI"
% "Somewhat Optimal Trajectory",0.7633,0.2513,0.00239,0.2708,1.256
% "WebArena Benchmark",0.7493,0.1696,9.89e-06,0.417,1.082
```

Created by data-to-paper (AI)

```
% "Looping Behavior",-2.642,0.1839,<1e-06,-3.002,-2.281
% "Annotator A",-0.9622,1.256,0.444,-3.424,1.5
% "Annotator B",0.0604,1.257,0.962,-2.404,2.525
% "Annotator C",-0.3491,1.256,0.781,-2.811,2.113
% "Annotator D",-0.5165,1.258,0.681,-2.982,1.949
% "Annotator E",-0.03205,1.299,0.98,-2.578,2.514
% "Annotator F",-0.6699,1.26,0.595,-3.14,1.8
% "Annotator G",0.2945,1.269,0.816,-2.192,2.781
% "Anthropic Claude 3.7 Model",-0.1552,0.204,0.447,-0.5551,0.2447
% "GPT-4o 11-20 Model",-0.05916,0.1963,0.763,-0.444,0.3256
% "Meta Llama 3.3 70B Model",-0.418,0.2453,0.0884,-0.8988,0.06282
% "Interaction: Somewhat Optimal * WebArena",-0.9628,0.3765,
    0.0106,-1.701,-0.2248
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Coefficient'])
%
% Confidence intervals for y-values were then plotted based on
    column: [('Lower CI', 'Upper CI')].
%
% P-values for y-values were taken from column: ['P-value'].
%
% These p-values were presented above the data points as stars (
    with significance threshold values indicated in the figure
    caption).
```